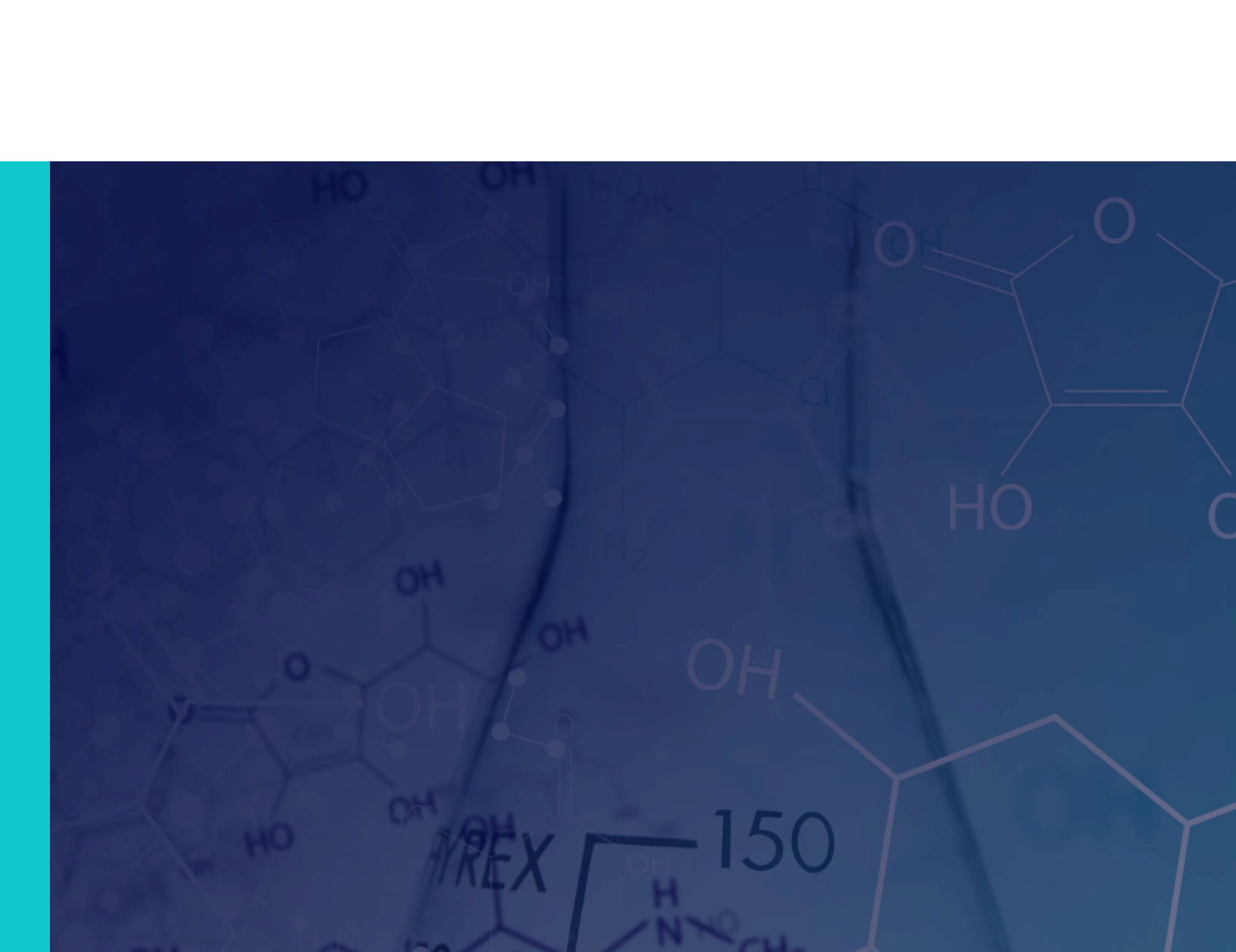




# U of A **CHEM 103**

Fall 2026, Chapter 1 Notes



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# 1. Unit 1. Quantum Model & Electron Configurations

## 1.1 Introduction to the Quantum Model

### 1.1.1 The Bohr Model Vs The Quantum Model

#### Bohr Model Vs Quantum Model

What was good about the Bohr model and theory?

✓ Could predict line spectra for hydrogen and other ions with **1 electron** (ex.  $\text{Li}^{2+}$ )

✗ But it couldn't predict line spectra for species with more than 1 electron

Couldn't explain how atoms were bound together

Didn't explain why only certain orbits were allowed

Tries to be too precise to describe where electrons are found

Bohr Theory was replaced by Quantum Mechanics!

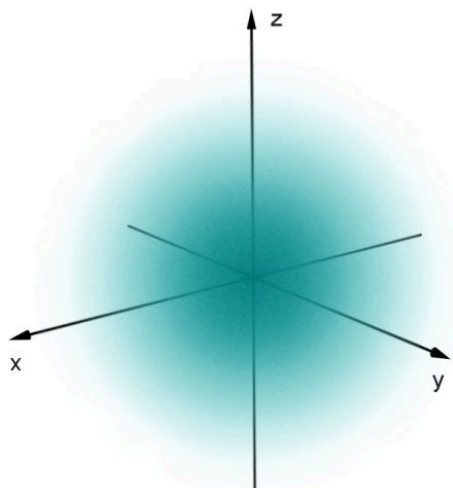
✓ Quantum model can predict atomic spectra for **many-electron atoms**

Quantum mechanics provides us with **probabilities where electrons are most likely to be found**

- Provides a 3D region where electrons are most likely to be found (instead of circular paths in Bohr model)

✓ Adds on to the Bohr model:

- Electrons are still only found at discrete energy levels with specific amounts of energy (quantized)
- In this model, **each electron has 4 quantum #s that are unique to that electron** and describe it.



Orbital s ( $\ell = 0$ ,  $m_\ell = 0$ )

### *Electron Density Diagram*

Photo by RJHall / CC BY

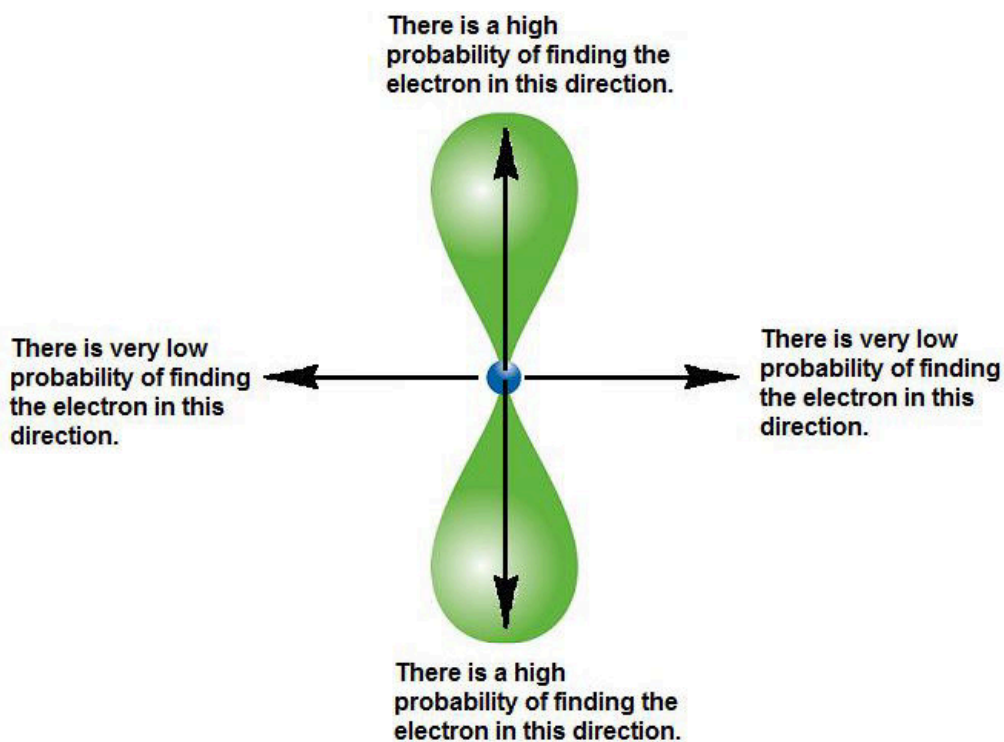


Photo by CK-12 Foundation / CC BY

## 1.2 The Quantum Model & Orbitals

### 1.2.1

### Quantum Numbers

Quantum numbers describe where electrons are positioned around atoms:

| Letter | Quantum Number           | Description         |
|--------|--------------------------|---------------------|
| $n$    | Principal                | Size                |
| $l$    | Orbital Angular Momentum | Shape               |
| $m_l$  | Magnetic                 | Orientation         |
| $m_s$  | Electronic Spin          | Electron Up or Down |

#### Rules:

$n$

- Can be **any positive integer**  
*Example:*  $n = (1, 2, 3, 4...)$
- As  $n$  increases, **energy and size** of shell \_\_\_\_\_

#### WIZE TIP

$n$  is also called the "principle quantum number"

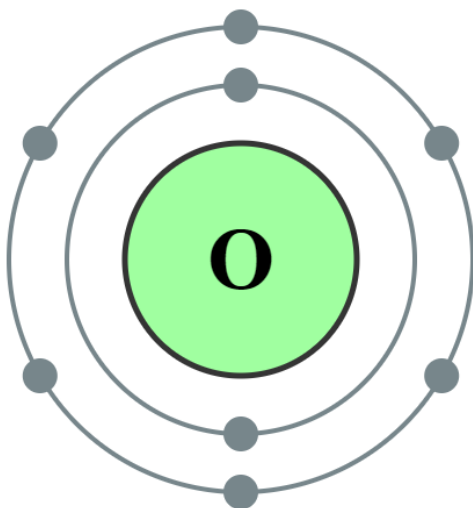


Photo by Greg Robson / CC BY

$l$

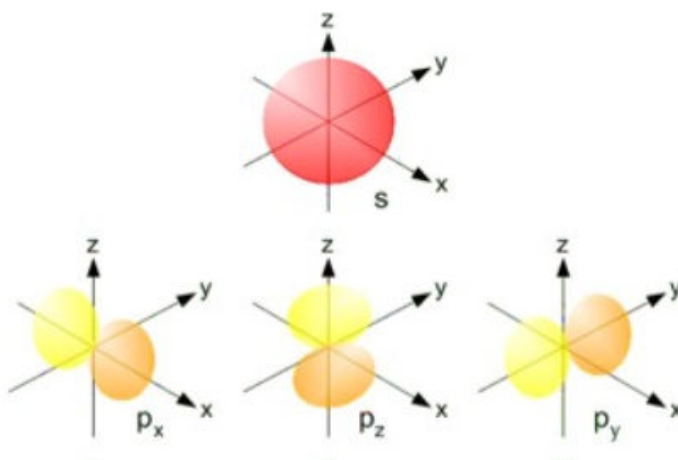
- Can be any **non-negative integer up to  $n-1$**

*Example:*  $l = (0, 1, 2, 3, \dots, n-1)$

- $l$  (**orbital shapes**) are described as:

| subshell | s | p | d | f |
|----------|---|---|---|---|
| $l$      | 0 | 1 | 2 | 3 |

Increasing energy  $\rightarrow$



- These shapes show us where an electron is most likely to be found
- There is a 90% chance in finding an electron somewhere inside the given shape

#### **WIZE CONCEPT**

**S orbitals are spheres** and with increasing "n" the sphere will get larger.  
There is one s orbital in a subshell.

**P subshells are dumbbell shaped.**

There are three p orbitals in a subshell, each orientated along an axis ( $p_x$ ,  $p_y$  and  $p_z$ )

---

$m_l$

- Can be any integer from **-l to +l**

*Example:*  $m_l = (-l, \dots, 0, \dots, l)$

- This quantum # **designates a specific orbital within a given shell**

*Example:* if  $n=2$  and  $l=1$ , we are looking at 2p.

- $m_l$  can be -1, 0, or +1 this designates each of the 2p orbitals: 2px, 2py, and 2pz

$m_s$

- **Spin** of an electron
- Can only be **+1/2 or -1/2**

## Quantum Numbers Summary

| Principle Quantum Number ( $n$ )    | Angular Momentum Quantum Number ( $\ell$ )                                                                   | Magnetic Quantum Number ( $m_\ell$ )                        | Electron-Spin Quantum Number ( $m_s$ )      |                                         |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------|-----------------------------------------|
| $n = 1, 2, 3, \dots$                | $\ell = 0, 1, 2, 3, \dots (n - 1)$                                                                           | $m_\ell = \text{integer values of } \ell \text{ to } -\ell$ | $m_s = \pm \frac{1}{2}$                     |                                         |
| $n \uparrow \rightarrow E \uparrow$ | $\ell = 0 \rightarrow s$<br>$\ell = 1 \rightarrow p$<br>$\ell = 2 \rightarrow d$<br>$\ell = 3 \rightarrow f$ | $2\ell + 1$ orbitals                                        | Only 2 electrons per orbital, <b>ALWAYS</b> |                                         |
| $n$ shell                           | $\ell$ subshell                                                                                              | Orbital                                                     | $m_\ell$                                    | Number of Orbitals per subshell = $n^2$ |
| 1                                   | 0                                                                                                            | 1s                                                          | 0                                           | 1                                       |
| 2                                   | 0                                                                                                            | 2s                                                          | 0                                           | 4                                       |
|                                     | 1                                                                                                            | 2p                                                          | +1, 0, -1                                   |                                         |
| 3                                   | 0                                                                                                            | 3s                                                          | 0                                           | 9                                       |
|                                     | 1                                                                                                            | 3p                                                          | +1, 0, -1                                   |                                         |
|                                     | 2                                                                                                            | 3d                                                          | +2, +1, 0, -1, -2                           |                                         |
| 4                                   | 0                                                                                                            | 4s                                                          | 0                                           | 16                                      |
|                                     | 1                                                                                                            | 4p                                                          | +1, 0, -1                                   |                                         |
|                                     | 2                                                                                                            | 4d                                                          | +2, +1, 0, -1, -2                           |                                         |
|                                     | 3                                                                                                            | 4f                                                          | +3, +2, +1, 0, -1, -2, -3                   |                                         |

## Example: Allowed Quantum Numbers

What are the allowed set of quantum numbers for the following orbitals?

a)  $6s$

b)  $4p$

c)  $3d$

## Example: Allowed Sets of Quantum Numbers

Which of the following sets of quantum numbers  $(n, l, m_l, m_s)$  are allowed and which are not allowed? For the sets of quantum orbitals that are not allowed, state why it is not allowed.

(i) (4, 0, 0, 0)

(ii) (3, 1, 2,  $-1/2$ )

(iii) (5, 3, 0,  $+1/2$ )

(iv) (4, 4, 3,  $-1/2$ )

## Example: Defining Orbitals from Quantum Numbers

Determine the atomic orbital described by the following sets of quantum numbers  $(n, l, m_l, m_s)$ .

(i)  $(2, 0, 0, -1/2)$

(ii)  $(4, 3, 0, +1/2)$

(iii)  $(5, 1, 1, -1/2)$

## Practice: Valid Quantum Numbers

Determine which of the following sets of quantum numbers ( $n$ ,  $l$ ,  $m_l$ ,  $m_s$ ) is valid for a  $3d$  orbital.

A)  $(3, 3, 2, +\frac{1}{2})$

☐

B)  $(4, 2, 1, -\frac{1}{2})$

☐

C)  $(3, 2, -2, 0)$

☐

D)  $(3, 2, -1, -\frac{1}{2})$

☐

E)  $(3, 2, 3, -\frac{1}{2})$

☐

F)  $(3, 1, 0, +\frac{1}{2})$

☐

# 1.3 Orbital Filling Diagrams

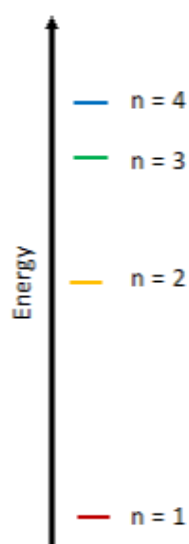
1.3.1

## Relative Energies of Atomic Orbitals

### One-Electron Species

- For a one-electron species like the hydrogen atom, there is no electron-electron repulsion. Therefore, all the subshells of the same  $n$  are **degenerate**.
- Orbitals at the **same energy level** are called **degenerate**.

Bohr Model of the Atom



Orbital Energy Level Diagram for the Hydrogen Atom

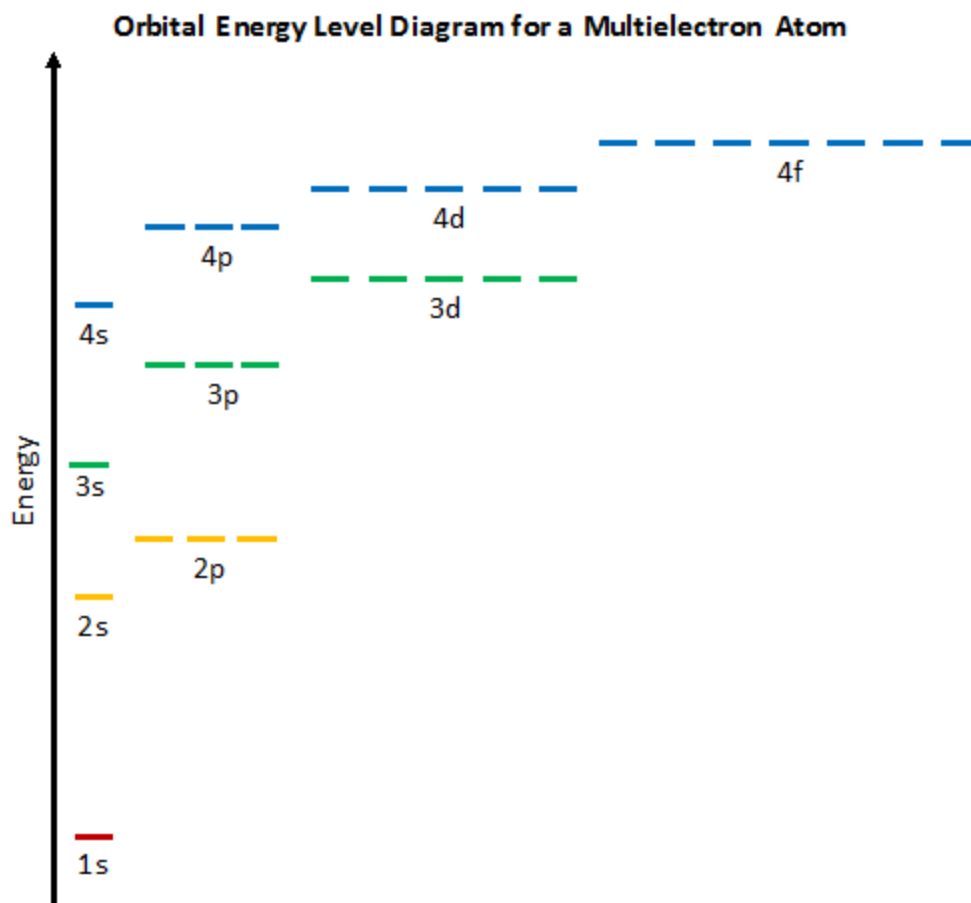


#### **i** WIZE TIP

It will be helpful to understand the difference between a **shell**, a **subshell**, and an **orbital**. We will label these in the diagram above!

## Multi-Electron Species

- With multi-electron species, there are interactions between electrons (electron-electron repulsion)
- The energy level diagram of the orbitals looks like this:





#### WIZE CONCEPT

- S subshells hold a maximum of \_\_\_\_\_ electrons
- P subshells hold a maximum of \_\_\_\_\_ electrons
- D subshells hold a maximum of \_\_\_\_\_ electrons
- F subshells hold a maximum of \_\_\_\_\_ electrons

The total number of orbitals in a given shell is given by :  $n^2$

*Example:*

**If  $n=2$ , what is the total number of orbitals in that shell?**

- \_\_\_\_\_, circle the \_\_\_\_\_ orbitals that have  $n=2$  above

**What if we wanted to know the maximum number of electrons we could have if  $n=2$ ?**

- We already found that total number of orbitals for this shell ( \_\_\_\_\_ )
- What is the maximum number of electrons we could have in each orbital? \_\_\_\_\_
- Therefore the maximum number of electrons possible when  $n=2$  is \_\_\_\_\_

## Rules for Orbital Filling

## 1) Aufbau Principle

Electrons will always occupy the **lowest available energy level first**.

|    |  |  |  |  |    |  |  |  |  |  |  |  |    |    |  |  |    |  |    |  |  |
|----|--|--|--|--|----|--|--|--|--|--|--|--|----|----|--|--|----|--|----|--|--|
| 1s |  |  |  |  |    |  |  |  |  |  |  |  |    |    |  |  | 1s |  |    |  |  |
| 2s |  |  |  |  |    |  |  |  |  |  |  |  |    |    |  |  |    |  | 2p |  |  |
| 3s |  |  |  |  |    |  |  |  |  |  |  |  |    |    |  |  |    |  | 3p |  |  |
| 4s |  |  |  |  | 3d |  |  |  |  |  |  |  | 4p |    |  |  |    |  |    |  |  |
| 5s |  |  |  |  | 4d |  |  |  |  |  |  |  | 5p |    |  |  |    |  |    |  |  |
| 6s |  |  |  |  | 5d |  |  |  |  |  |  |  | 6p |    |  |  |    |  |    |  |  |
| 7s |  |  |  |  | 6d |  |  |  |  |  |  |  | 7p |    |  |  |    |  |    |  |  |
|    |  |  |  |  |    |  |  |  |  |  |  |  |    |    |  |  |    |  |    |  |  |
|    |  |  |  |  |    |  |  |  |  |  |  |  |    | 4f |  |  |    |  |    |  |  |
|    |  |  |  |  |    |  |  |  |  |  |  |  |    | 5f |  |  |    |  |    |  |  |

Diagram illustrating the filling order of atomic orbitals (Aufbau principle):

- 1s
- 2s 2p
- 3s 3p 3d
- 4s 4p 4d 4f
- 5s 5p 5d 5f ...
- 6s 6p 6d ...

**! WATCH OUT!**

**Memorize the order of orbital filling (or just familiarize yourself enough with the periodic table to know the order!):** 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p...

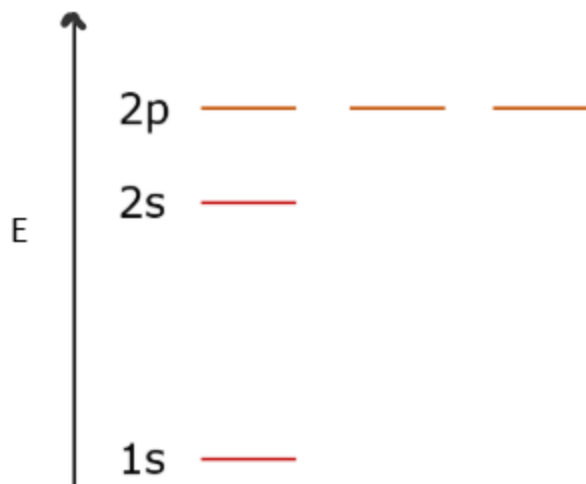
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## 2) Hund's Rule

Due to electron-electron repulsion, electrons will **fill orbitals of the same energy singly before pairing up**

Electrons don't want to be next to each other unless they have to be!

*Example:* Fill out the following orbital diagram for C



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### 3) Pauli Exclusion Principle

No two electrons in an atom will have the same set of **4 quantum numbers**.

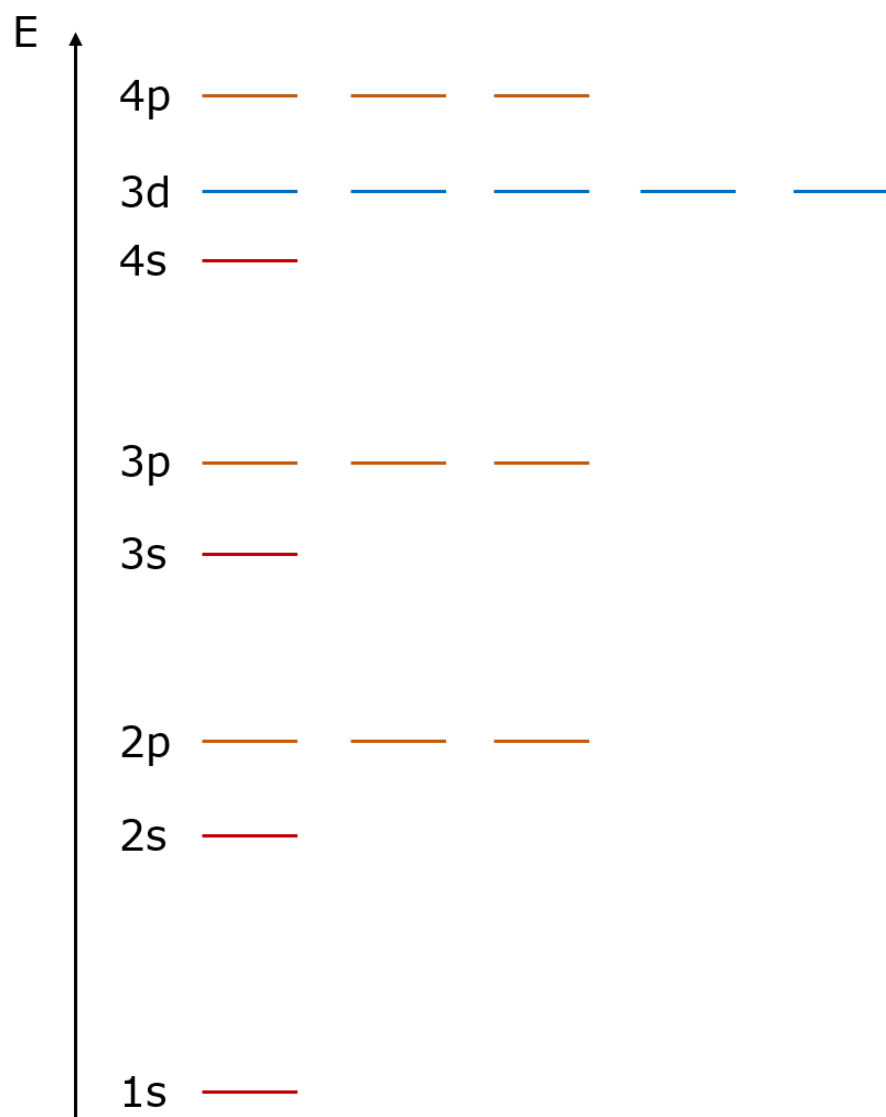
| Quantum Number                  | Values       | Interpretation                                                                                                                    |
|---------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $m_s$ (the spin quantum number) | -1/2 or +1/2 | An electron behaves like a magnet that has one of two possible orientations, aligned either with the magnetic field or against it |

## Example: Orbital Filling Diagrams

Draw the orbital diagram for oxygen.

## Example: Orbital Filling Diagrams

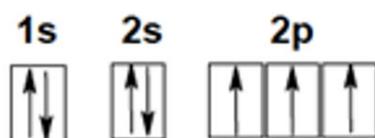
Complete the atomic orbital diagram below for a neutral calcium atom.



## Practice: Orbital Filling Diagrams

Which of the orbital diagrams gives the correct electron configuration for an atom of boron, in the ground state?

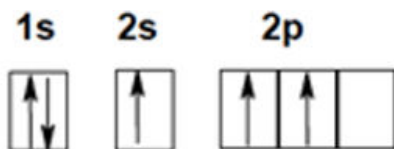
A)

☐

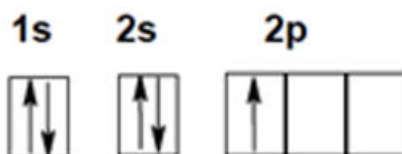
B)

☐

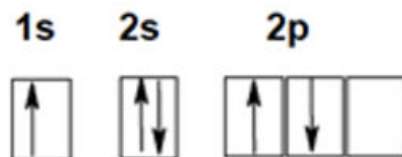
C)

☐

D)

☐

E)

☐

## Practice: Number of Orbitals

How many different atomic orbitals exist where  $n = 3$ ?

3

☐

6

☐

9

☐

12, which

☐

18

☐

# 1.4 Electronic Configuration for Atoms

## 1.4.1

### Electron Configurations of Atoms

| Group →     | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Ms | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides | 57<br>La | 58<br>Ce | 59<br>Pr | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |           |           |
| Actinides   | 89<br>Ac | 90<br>Th | 91<br>Pa | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |           |           |

- Label blocks
- describe the n number for each block

### Simple Electron Configurations

Li:

C:

Ne:

| Group →     | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Ms | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides | 57<br>La | 58<br>Ce | 59<br>Pr | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |           |           |
| Actinides   | 89<br>Ac | 90<br>Th | 91<br>Pa | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |           |           |

### Shorthand Notations

Write the name of the **previous noble gas** and then fill in the rest of the electron configuration.

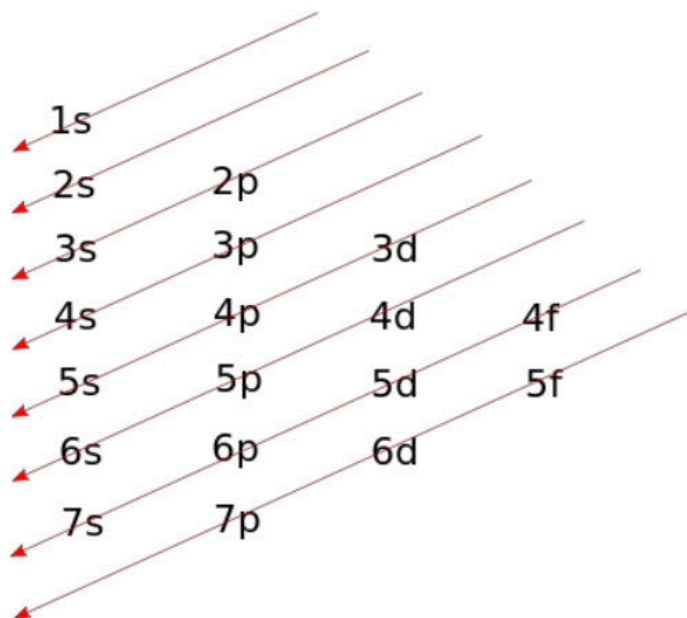
**O:**

**Fe:**

**Ge:**

## Electron Configurations Cheatsheet

|                             |                             |                           |                            |                             |                             |                            |                           |                            |                            |                            |                            |                            |                            |                            |                            |                            |                            |                           |                         |                          |                           |  |
|-----------------------------|-----------------------------|---------------------------|----------------------------|-----------------------------|-----------------------------|----------------------------|---------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|---------------------------|-------------------------|--------------------------|---------------------------|--|
| 1<br>H<br>1.008<br>←1s→     |                             |                           |                            |                             |                             |                            |                           |                            |                            |                            |                            |                            |                            |                            |                            |                            | 2<br>He<br>4.003           |                           |                         |                          |                           |  |
| 3<br>Li<br>6.94<br>←2s→     | 4<br>Be<br>9.012            |                           |                            |                             |                             |                            |                           |                            |                            |                            |                            |                            |                            |                            |                            | 5<br>B<br>10.81<br>2p→     | 6<br>C<br>12.011<br>2p→    | 7<br>N<br>14.007<br>2p→   | 8<br>O<br>15.999<br>2p→ | 9<br>F<br>18.998<br>2p→  | 10<br>Ne<br>20.180<br>2p→ |  |
| 11<br>Na<br>22.990<br>←3s→  | 12<br>Mg<br>24.305<br>←3s→  |                           |                            |                             |                             |                            |                           |                            |                            |                            |                            |                            |                            |                            |                            | 13<br>Al<br>26.982<br>3p→  | 14<br>Si<br>28.085<br>3p→  | 15<br>P<br>30.974<br>3p→  | 16<br>S<br>32.06<br>3p→ | 17<br>Cl<br>35.45<br>3p→ | 18<br>Ar<br>39.948<br>3p→ |  |
| 19<br>K<br>39.098<br>←4s→   | 20<br>Ca<br>40.078<br>←4s→  | 21<br>Sc<br>44.956<br>3d→ | 22<br>Ti<br>47.867<br>3d→  | 23<br>V<br>50.942<br>3d→    | 24<br>Cr<br>51.996<br>3d→   | 25<br>Mn<br>54.938<br>3d→  | 26<br>Fe<br>55.845<br>3d→ | 27<br>Co<br>58.933<br>3d→  | 28<br>Ni<br>58.693<br>3d→  | 29<br>Cu<br>63.546<br>3d→  | 30<br>Zn<br>65.38<br>3d→   | 31<br>Ga<br>69.723<br>4p→  | 32<br>Ge<br>72.630<br>4p→  | 33<br>As<br>74.922<br>4p→  | 34<br>Se<br>78.97<br>4p→   | 35<br>Br<br>79.904<br>4p→  | 36<br>Kr<br>83.798<br>4p→  |                           |                         |                          |                           |  |
| 37<br>Rb<br>85.468<br>←5s→  | 38<br>Sr<br>87.62<br>←5s→   | 39<br>Y<br>88.906<br>4d→  | 40<br>Zr<br>91.224<br>4d→  | 41<br>Nb<br>92.906<br>4d→   | 42<br>Mo<br>95.95<br>4d→    | 43<br>Tc<br>[97]<br>4d→    | 44<br>Ru<br>101.07<br>4d→ | 45<br>Rh<br>102.906<br>4d→ | 46<br>Pd<br>106.42<br>4d→  | 47<br>Ag<br>107.868<br>4d→ | 48<br>Cd<br>112.414<br>4d→ | 49<br>In<br>114.818<br>5p→ | 50<br>Sn<br>118.710<br>5p→ | 51<br>Sb<br>121.760<br>5p→ | 52<br>Te<br>127.60<br>5p→  | 53<br>I<br>126.905<br>5p→  | 54<br>Xe<br>131.29<br>5p→  |                           |                         |                          |                           |  |
| 55<br>Cs<br>132.905<br>←6s→ | 56<br>Ba<br>137.327<br>←6s→ | *<br>57 - 70              | 71<br>La<br>138.905<br>5d→ | 72<br>Ce<br>140.116<br>5d→  | 73<br>Pr<br>140.908<br>5d→  | 74<br>Nd<br>144.242<br>5d→ | 75<br>Pm<br>[145]<br>5d→  | 76<br>Sm<br>150.36<br>5d→  | 77<br>Eu<br>151.964<br>5d→ | 78<br>Gd<br>157.25<br>5d→  | 79<br>Tb<br>158.925<br>5d→ | 80<br>Dy<br>162.500<br>5d→ | 81<br>Ho<br>164.930<br>5d→ | 82<br>Er<br>167.259<br>5d→ | 83<br>Tm<br>168.934<br>5d→ | 84<br>Yb<br>173.045<br>5d→ | 85<br>Lu<br>174.967<br>5d→ |                           |                         |                          |                           |  |
| 87<br>Fr<br>[223]<br>←7s→   | 88<br>Ra<br>[226]<br>←7s→   | **<br>89 - 102            | 103<br>Ac<br>[227]<br>6d→  | 104<br>Th<br>232.038<br>6d→ | 105<br>Pa<br>231.036<br>6d→ | 106<br>U<br>238.029<br>6d→ | 107<br>Np<br>[237]<br>6d→ | 108<br>Pu<br>[244]<br>6d→  | 109<br>Am<br>[243]<br>6d→  | 110<br>Cm<br>[247]<br>6d→  | 111<br>Bk<br>[247]<br>6d→  | 112<br>Cf<br>[251]<br>6d→  | 113<br>Es<br>[252]<br>6d→  | 114<br>Fm<br>[257]<br>6d→  | 115<br>Md<br>[258]<br>6d→  | 116<br>Lv<br>[260]<br>6d→  | 117<br>Ts<br>[261]<br>6d→  | 118<br>Og<br>[264]<br>6d→ |                         |                          |                           |  |
| *Lanthanide series          |                             | 4f→                       | 57<br>La<br>138.905        | 58<br>Ce<br>140.116         | 59<br>Pr<br>140.908         | 60<br>Nd<br>144.242        | 61<br>Pm<br>[145]         | 62<br>Sm<br>150.36         | 63<br>Eu<br>151.964        | 64<br>Gd<br>157.25         | 65<br>Tb<br>158.925        | 66<br>Dy<br>162.500        | 67<br>Ho<br>164.930        | 68<br>Er<br>167.259        | 69<br>Tm<br>168.934        | 70<br>Yb<br>173.045        |                            |                           |                         |                          |                           |  |
| **Actinide series           |                             | 5f→                       | 89<br>Ac<br>[227]          | 90<br>Th<br>232.038         | 91<br>Pa<br>231.036         | 92<br>U<br>238.029         | 93<br>Np<br>[237]         | 94<br>Pu<br>[244]          | 95<br>Am<br>[243]          | 96<br>Cm<br>[247]          | 97<br>Bk<br>[247]          | 98<br>Cf<br>[251]          | 99<br>Es<br>[252]          | 100<br>Fm<br>[257]         | 101<br>Md<br>[258]         | 102<br>No<br>[259]         |                            |                           |                         |                          |                           |  |



## Electron Configuration Exceptions

Transition metals have their valence electrons in the d subshell.

It is a lot **more favourable** (and stable) to have **d shell either half full or totally full instead of just partially full**.

|                  | Cr:                                                                                                                                  |                                                                                                                                         | Cu: |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Expected:</b> | <br>4s 3d<br>[Ar] 4s <sup>2</sup> 3d <sup>4</sup>   | <br>4s 3d<br>[Ar] 4s <sup>2</sup> 3d <sup>9</sup>    |     |
| <b>Seen:</b>     | <br>4s 3d<br>[Ar] 4s <sup>1</sup> 3d <sup>5</sup> | <br>4s 3d<br>[Ar] 4s <sup>1</sup> 3d <sup>10</sup> |     |

### **i** WIZE TIP

The other exceptions that follow this same trend are: Mo, Ag, and Au! (Note that they are in the same groups that Cr and Cu are in!)

## Example: Writing Out Electron Configurations

### Part 1:

Write the full electronic configurations for the following elements:

**P:**

**Se:**

### Part 2:

Write the short electron configurations for the following elements:

**Zn:**

**Zr:**

| Group→      | 1                                  | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|------------------------------------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    | The Periodic Table of the Elements |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H                             |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li                            | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na                           | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K                            | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb                           | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs                           | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr                           | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Mc | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides |                                    |          | 57<br>La | 58<br>Ce  | 59<br>Pr  | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |
| Actinides   |                                    |          | 89<br>Ac | 90<br>Th  | 91<br>Pa  | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |

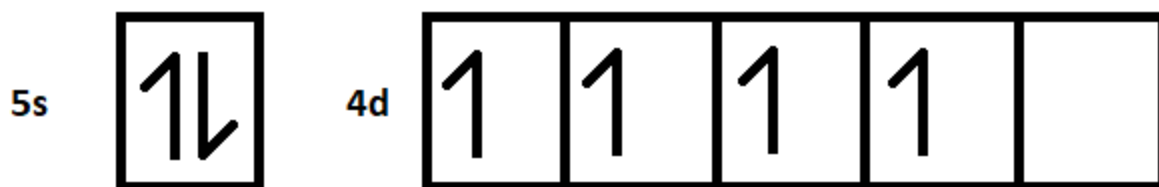
## Example: Condensed Electron Configuration

What is the condensed electronic configuration for Mo?

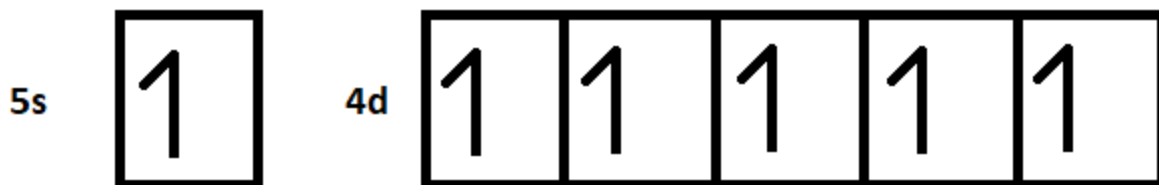
| Group→      | 1                                  | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|------------------------------------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    | The Periodic Table of the Elements |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H                             |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li                            | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na                           | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K                            | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb                           | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs                           | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr                           | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Mc | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides |                                    |          | 57<br>La | 58<br>Ce  | 59<br>Pr  | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |
| Actinides   |                                    |          | 89<br>Ac | 90<br>Th  | 91<br>Pa  | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |

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We ignore everything before the previous noble gas (Kr), which means we are only working with the 5s and 4d blocks. Let's fill them up:



But there is something special about Mo. It is in group 6. This group tends to fill (or in this case half-fill) the d-block at the expense of the s block.



Therefore, the electronic configuration is  $[\text{Kr}]5s^14d^5$ .

## Practice: Identify the Neutral Element

What neutral element is represented in the example below?

$1s^2 2s^2 2p^6 3s^2 3p^5$  became  $[Ne] 3s^2 3p^5$

Al

☐

Ne

☐

P

☐

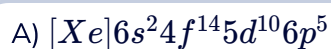
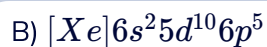
Cl

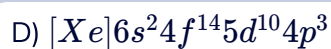
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## Practice: Electron Configuration

The ground state electronic configuration for At is:

| Group →     | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Mc | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides |          |          | 57<br>La | 58<br>Ce  | 59<br>Pr  | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |
| Actinides   |          |          | 89<br>Ac | 90<br>Th  | 91<br>Pa  | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |


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## 1.4.8 Practice

### Practice: Impossible Electron Configuration

Which of the following electronic configurations is not possible?

Group → 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

↓ Period

**The Periodic Table of the Elements**

|             |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
|-------------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 1<br>H      |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 3<br>Li     | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 11<br>Na    | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 19<br>K     | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 37<br>Rb    | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 55<br>Cs    | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 87<br>Fr    | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Mc | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides |          | 57<br>La | 58<br>Ce  | 59<br>Pr  | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |
| Actinides   |          | 89<br>Ac | 90<br>Th  | 91<br>Pa  | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |

[Ar]4s<sup>2</sup>4d<sup>6</sup>

☐

1s<sup>2</sup>2s<sup>2</sup>2p<sup>6</sup>3s<sup>2</sup>3p<sup>6</sup>4s<sup>1</sup>

☐

[Kr]5s<sup>1</sup>

☐

1s<sup>2</sup>2s<sup>2</sup>2p<sup>6</sup>3s<sup>2</sup>3p<sup>6</sup>4s<sup>2</sup>3d<sup>8</sup>

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# 1.5 Electron Configurations for Ions

## 1.5.1

### Electron Configuration of Ions of Main Group Elements

We will now look at how to write the electron configurations for ions!

**Writing the electron configurations for anions (negative ions)** are simple.

1. Write out the electron configuration for the **neutral element**
2. Fill in the desired number of electrons by **adding electron(s) to the highest energy subshell**

**Writing the electron configurations for cations (positive ions)** usually confuses students more, but there is a simple thing to remember to get these questions right every time!

1. Write out the electron configuration for the **neutral element**
2. Remove the desired number of electrons by **first removing electrons from the highest n level and highest energy subshell!**

#### WIZE CONCEPT

**The higher the n level, the higher the energy.**

**Example:** 2s has a higher energy than 1s

In order of **increasing energy for the subshells**, we write:

**$s < p < d < f$**

**Example:** 2p has a higher energy than 2s

---

F:  $1s^2 2s^2 2p^5$

[He]  $2s^2 2p^5$



---

F<sup>-</sup>:  $1s^2 2s^2 2p^6$

[Ne]



What is the electron configuration for F<sup>+</sup>?

**i WIZE TIP**

The most stable ions of atoms are **isoelectronic** with the noble gases or have filled shells. **Isoelectronic species** have the same number of electrons and the same electron configuration. The term "isoelectronic" is commonly seen on exams.

**Example:** F<sup>-</sup> is isoelectronic with Ne!

## Electron Configuration of Ions of Transition Elements

The same rules that we just learned apply for transition elements as well. Note that valence electrons are only removed, never core electrons.

### ! WATCH OUT!

Transition metals can lose both “n” and “n-1” valence electrons, but “n” electrons are always lost first! Let's take a look!

**Fe:** [Ar] 4s<sup>2</sup>3d<sup>6</sup>



**Fe<sup>1+</sup>:** [Ar] 4s<sup>1</sup>3d<sup>6</sup>



**Fe<sup>2+</sup>:** [Ar] 3d<sup>6</sup>



**Fe<sup>3+</sup>:** [Ar] 3d<sup>5</sup>



### i WIZE TIP

Always write out the **electron configuration for the neutral atom first**.  
Then write out the **electron configuration for the charged element**.

This will help you avoid mistakes on the exam!

## Example: Isoelectronic Species

Two atoms or ions are said to be isoelectronic if the electron configuration is the same for both species. Which of the following pairs are isoelectronic?

| Group →     | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Ms | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides |          |          | 57<br>La | 58<br>Ce  | 59<br>Pr  | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |
| Actinides   |          |          | 89<br>Ac | 90<br>Th  | 91<br>Pa  | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |

i)  $\text{N}^+$  and C

ii) C and  $\text{B}^+$

---

iii)  $\text{Cl}^-$  and Ar

## Example: Writing Electron Configurations of Ions

Write the electronic configuration for the following species:

| Group →     | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Ms | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides | 57<br>La | 58<br>Ce | 59<br>Pr | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |           |           |
| Actinides   | 89<br>Ac | 90<br>Th | 91<br>Pa | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |           |           |

a)  $\text{N}^+$

b)  $\text{Ni}^{2+}$

---

c)  $\text{Ge}^-$

## Practice: Electron Configurations of Ions

For which of the atom/ions below, is the given electron configuration correct? (select all that apply)

|      | Species          | Electron Configurations              |
|------|------------------|--------------------------------------|
| i.   | Cu               | [Ar] 4s <sup>2</sup> 3d <sup>9</sup> |
| ii.  | Ti <sup>2+</sup> | [Ar] 4s <sup>2</sup>                 |
| iii. | As <sup>5+</sup> | [Ar] 4s <sup>2</sup> 3d <sup>8</sup> |

i.

☐

ii.

☐

iii.

☐

None of the above

☐

---

# 1.6 Diamagnetic vs Paramagnetic Electron Configurations

1.6.1

## Diamagnetic vs Paramagnetic

You may see these terms come up when talking about electron configurations.

**Diamagnetic** - all electrons are **paired**

**Paramagnetic** - all electrons are **not paired**

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Are the following electron configurations diamagnetic or paramagnetic?

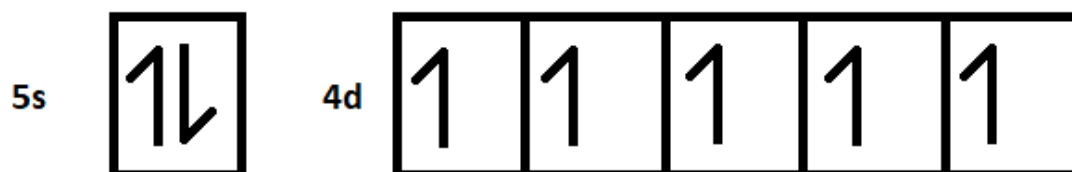
Cd

[Kr] 5s<sup>2</sup> 4d<sup>10</sup>



Tc

[Kr] 5s<sup>2</sup> 4d<sup>5</sup>



|             |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
|-------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Group →     | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
| ↓ Period    |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Ms | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides | 57<br>La | 58<br>Ce | 59<br>Pr | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |           |           |
| Actinides   | 89<br>Ac | 90<br>Th | 91<br>Pa | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |           |           |

Consider the electron configuration for Na

Is it diamagnetic or paramagnetic? \_\_\_\_\_

Consider the electron configuration for Si

Is it diamagnetic or paramagnetic? \_\_\_\_\_

Consider the electron configuration for Be

---

Is it diamagnetic or paramagnetic? \_\_\_\_\_

---

 **WIZE CONCEPT**

**Diamagnetic**-all electrons are **paired**. If something is diamagnetic it will also be repelled from externally produced magnetic fields.

**Paramagnetic**-One or more electrons are left **unpaired**. If something is paramagnetic it will be attracted to an externally produced magnetic field.

## Example: Paramagnetic and Diamagnetic Species

Write the electronic configuration of the following species, and label them as **D** diamagnetic or **P** paramagnetic

| Group →     | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Ms | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides | 57<br>La | 58<br>Ce | 59<br>Pr | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |           |           |
| Actinides   | 89<br>Ac | 90<br>Th | 91<br>Pa | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |           |           |

a)  $\text{Sc}^{2+}$

b)  $\text{Cr}^{2+}$

---

c)  $\text{Mn}^{7+}$

## Practice: Diamagnetic

Which of the following are diamagnetic? (select all that apply)

| Group →     | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |
|-------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ↓ Period    |          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |
| 1           | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |
| 2           | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |
| 3           | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |
| 4           | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |
| 5           | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |
| 6           | 55<br>Cs | 56<br>Ba |          | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7           | 87<br>Fr | 88<br>Ra |          | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Ms | 116<br>Lv | 117<br>Ts | 118<br>Og |
| Lanthanides |          |          | 57<br>La | 58<br>Ce  | 59<br>Pr  | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |
| Actinides   |          |          | 89<br>Ac | 90<br>Th  | 91<br>Pa  | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |

Cr

☐

Co

☐

Ru

☐

None of the above

☐

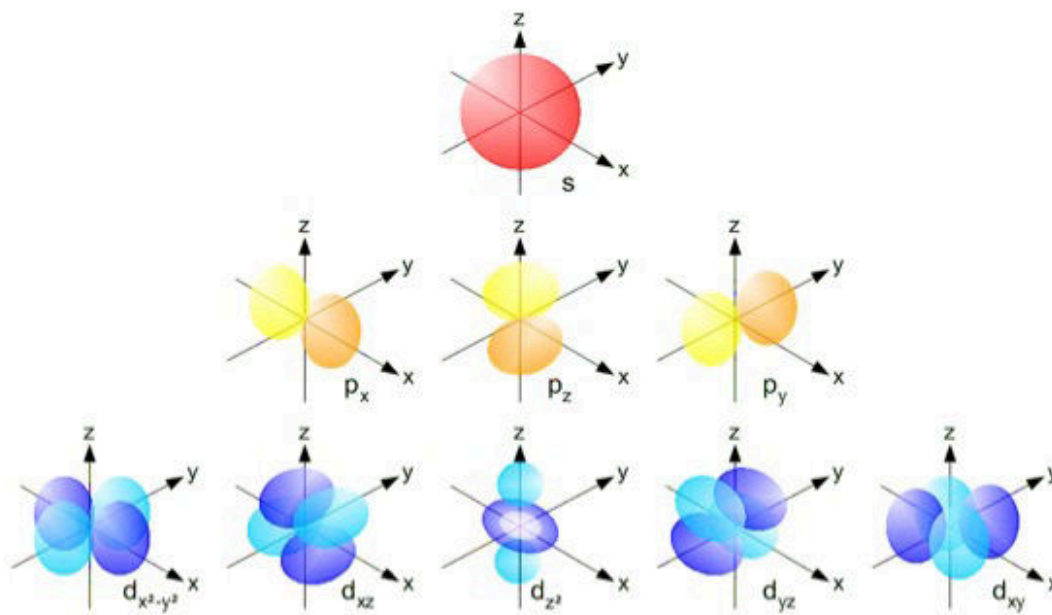
**View Solutions on Wizeprep.com**

Solutions to these questions, as well as step-by-step breakdowns of the answers at:

# 1.7 Shapes of Atomic Orbitals

## 1.7.1

### Shapes (aka Boundary Diagrams) of Atomic Orbitals



- There is **one s orbital** in a subshell, which is a **sphere**
- There are **three p orbitals** in a subshell, each orientated along an axis ( **$p_x$ ,  $p_y$  and  $p_z$** )



- 
- There are **five d orbitals** in a subshell. The shapes of the d orbitals are more complicated than their s and p counterparts.
    - Three orbitals look like 3D cloverleaves, each lying in a plane with the lobes pointed between the axes ( $d_{xy}$ ,  $d_{xz}$  and  $d_{yz}$ ).
    - A fourth orbital is also a cloverleaf, but its lobes point along the axes ( $d_{x^2-y^2}$ ).
    - The fifth orbital looks quite different with its major lobes pointing along the z- axis, but there is also a “doughnut” of electron density in the xy plane ( $d_{z^2}$ )

## Things to Keep in Mind:

- As the **energy level increases**, so does the **size of the orbitals**

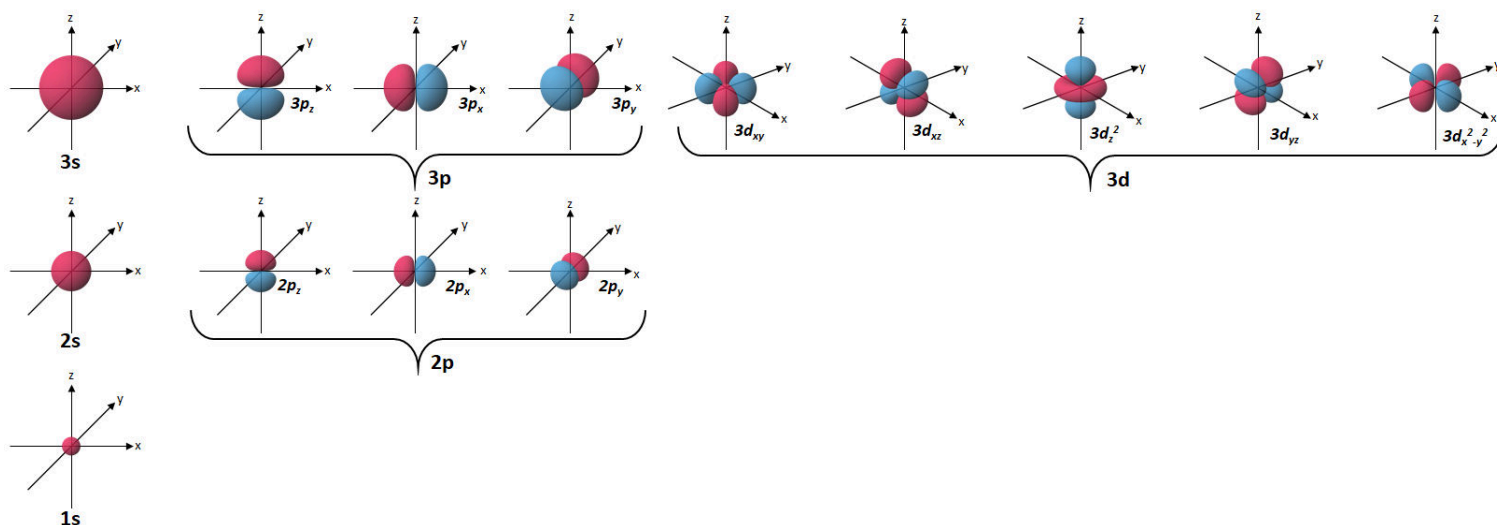
*Example:* A 2s orbital is higher in energy and larger than a 1s orbital

- As the **energy level increases**, so does the **number of orbitals**

*Example:* when  $n=1$  (lower energy level), we see 1 orbital, whereas when  $n=2$  (higher energy level), there are 4 possible orbitals

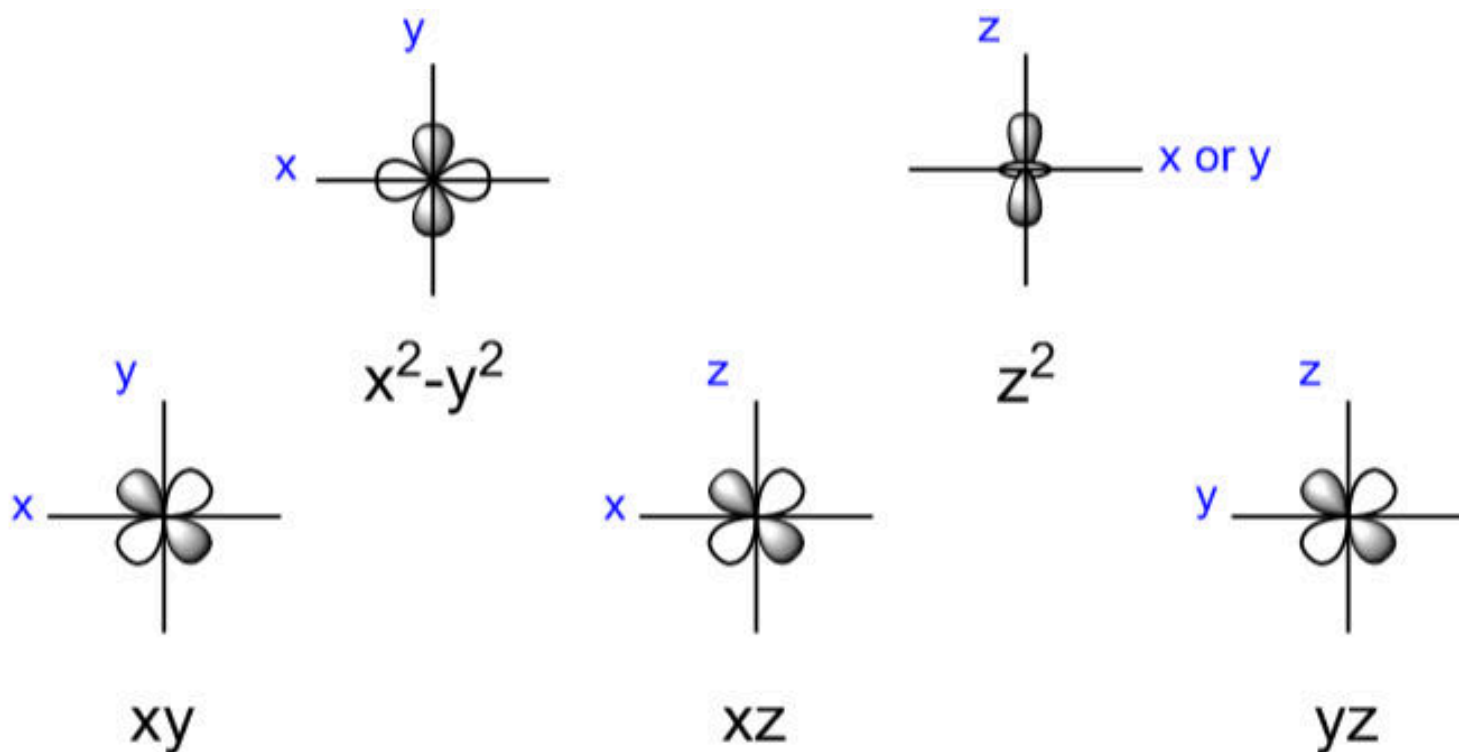
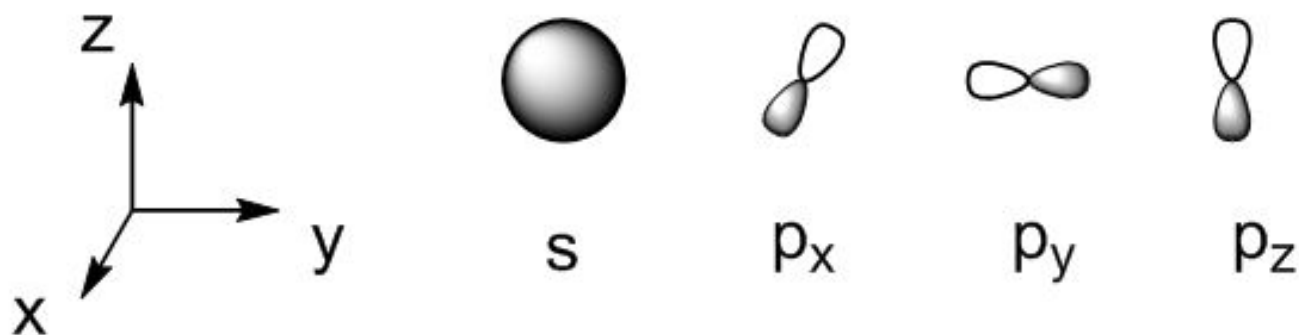
- As the **number of orbitals increases**, so does their **complexity**

*Example:* d orbitals are more complex than p orbitals



## Drawing Atomic Orbitals

Shown below are the s, p and d orbitals:



**i WIZE TIP**

Pay attention to whether the lobes are ON an axis or BETWEEN axes!

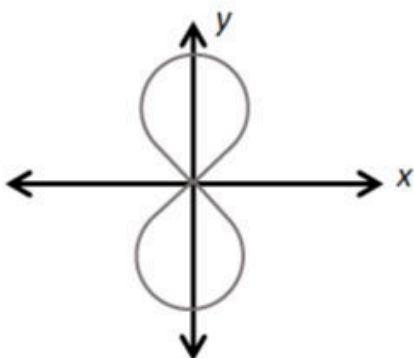
Also, notice how every other lobe is colored in gray. This gray and white coloring is indicating a "positive" lobe and a "negative" lobe. If you were asked to draw one of these orbitals on an exam, you would be expected to color in the lobes as well (or indicate which are + and which are -).

- The angular momentum quantum number,  $l$ , gives us information about the shape of the orbital.
- **Note:** There are 5 d-orbitals and 5 allowed values for  $m_l$ . Similarly for s and p the number of orbitals is determined by the # of allowed values of  $m_l$ .
- **Note:** Each  $m_l$  is not associated with a particular orbital (ex.  $d_{xy}$  vs  $d_{xz}$ ). Which means given a set of quantum numbers  $(n, l, m_l)$  one cannot determine which direction the orbital will be pointing

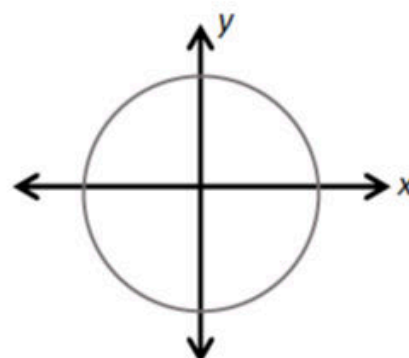
## Example: Orbitals

The sketches below show possible orbitals for the electron in a hydrogen atom. Which orbital would have the *lowest* energy?

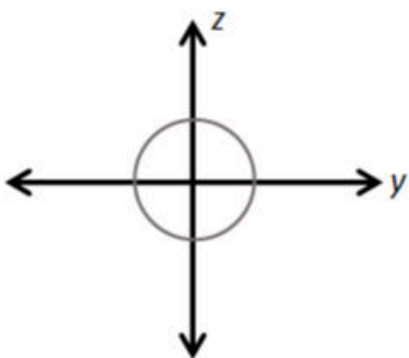
a.



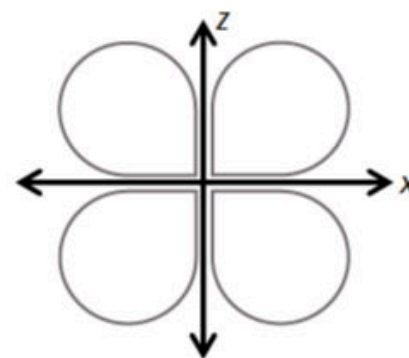
b.



c.

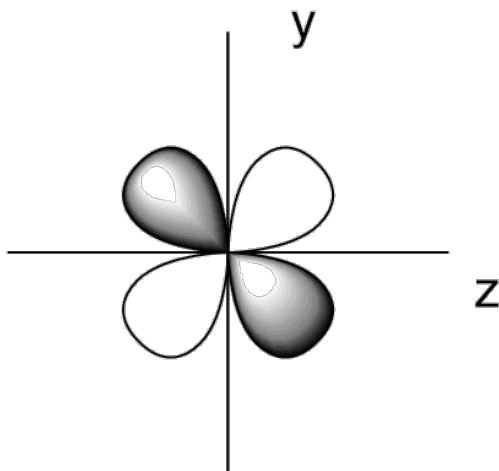


d.



## Practice: Identify the Orbital

Identify the orbital shown below,



a)  $2p_y$

☐

b)  $3d_{yz}$

☐

c)  $3p_x$

☐

d)  $3d_{xy}$

☐

e)  $3d_{x^2-y^2}$

☐

## 1.8 Radial and Angular Nodes

### 1.8.1

### Angular and Radial Nodes

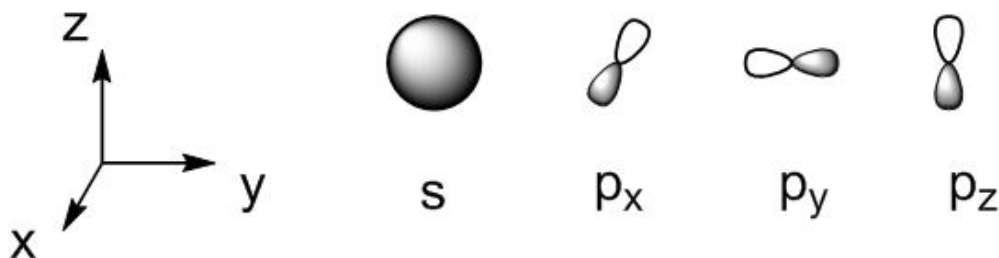
$$\text{Total number of nodes} = n - 1$$

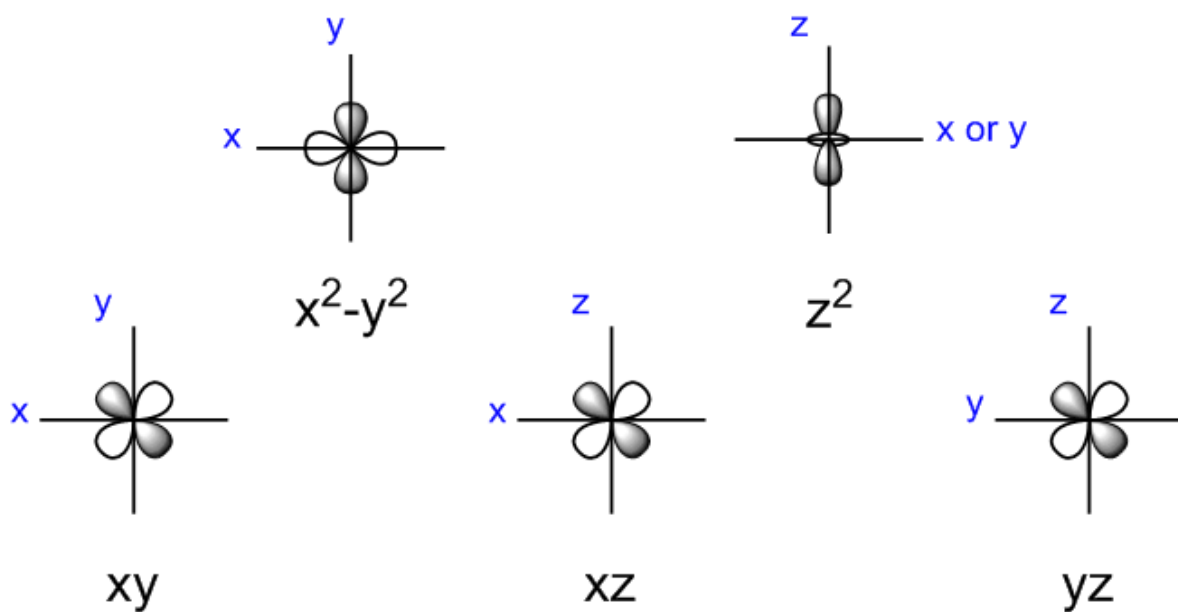
$$\text{Number of angular nodes} = l$$

$$\text{Number of radial nodes} = n - 1 - l$$

- There are two types of nodes: radial and angular.
- The number of angular nodes depends on the quantum number  $l$ .

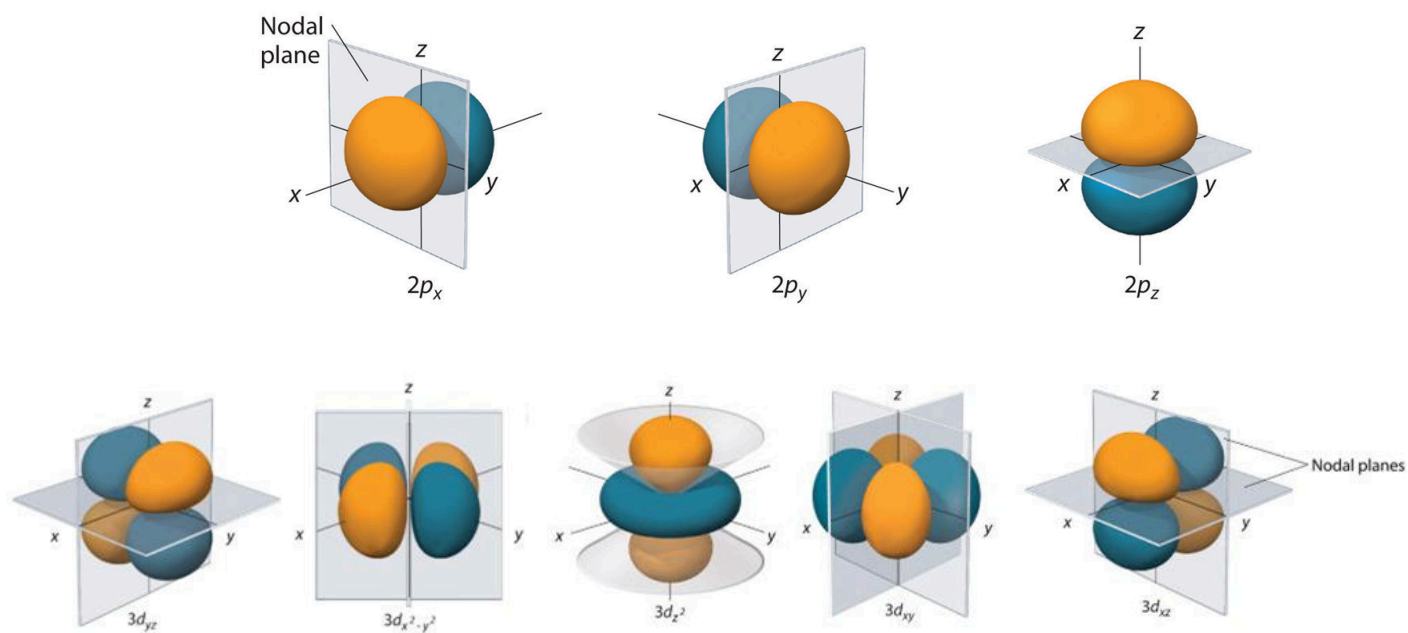
This time let's draw in the angular nodes for the 2p and 3d orbitals:





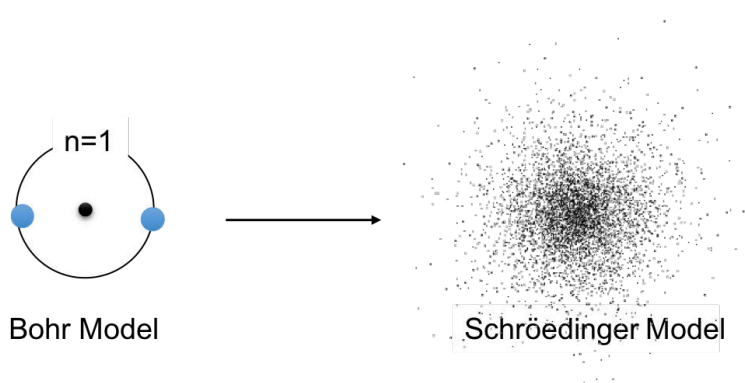
### Another View of the Nodes

- **Nodes** are regions in space where the probability of finding an electron is zero.
- In p orbitals and d orbitals, we have **planar nodes** between the lobes of electron densities.
- We will take a look at s orbitals and their nodes soon...

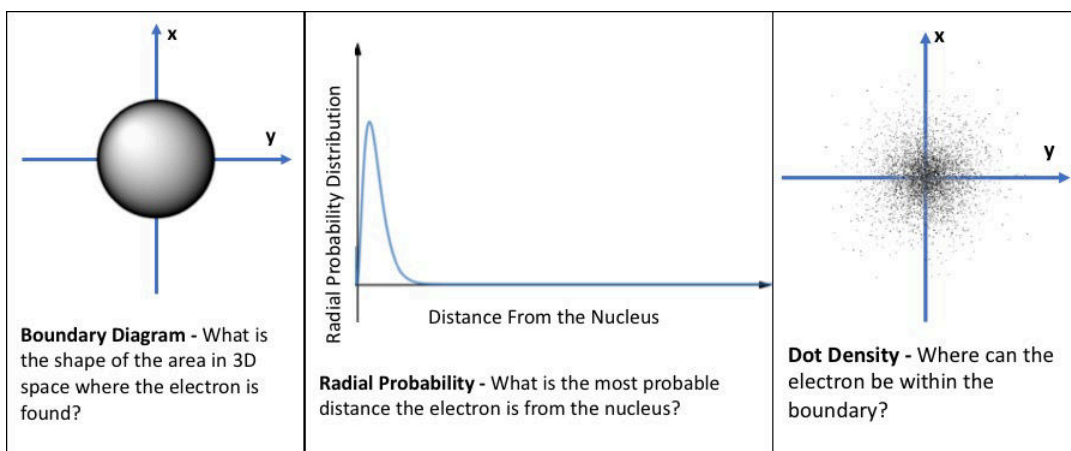


## Wave Properties of Electrons

- Electrons have both particle and wave-like properties. Within an atom, electrons are standing waves, oscillating around the nucleus. As with all waves, there are nodes and phases.
- Orbital:** mathematically derived region of probability in 3D space where an electron “may be” found



## Representations of Atomic Orbitals



## Plotting the Radial Probability

The wavefunction is a mathematical function and it can be graphed like any other function.

### WIZE TIP

We will not need to know any exact values of these graphs, just their **general shape**.

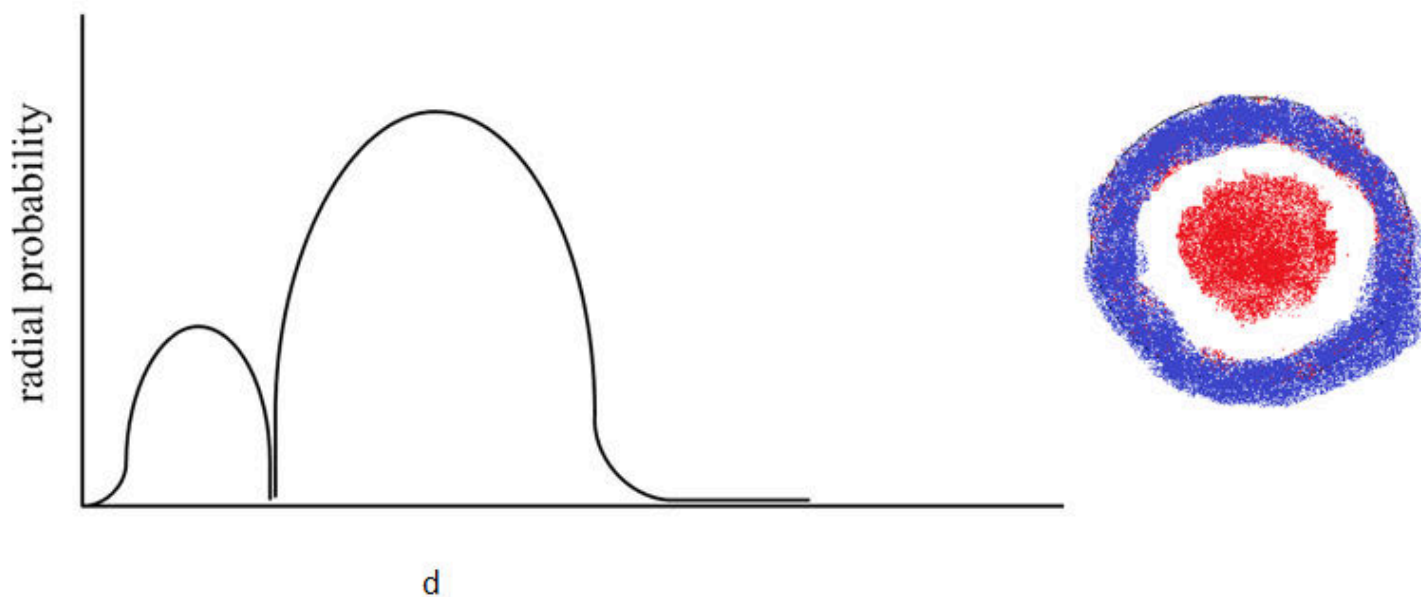
The only thing you will have need to “memorize” to do these problems is how to **draw each orbital**.

Although the wavefunction has no physical meaning, the square of the wavefunction is related to the probability of finding an electron at a given point in space.

### WIZE CONCEPT

- Each **radial node** will be indicated by a **zero point on a radial probability graph**
- **Each graph begins at zero** (no probability of finding the electron in the nucleus)
- **Each graph trails off to infinity** (no “edge” of an orbital”)

*Example:* 2s Orbital



**! WATCH OUT!**

The point where the **distance from the nucleus = 0** is **NOT a radial node!**  
This is a common mistake students make on exams.

Aside from that point, **every other point that touches the x axis represents a radial node!**

**i WIZE TIP**

The most important piece of information we can get from the **radial probability plot** is the **number of radial nodes!**

## Example: Calculate the Number of Radial and Angular Nodes

How many radial and angular nodes are present in the following orbitals?

a) 7p

b) 3d

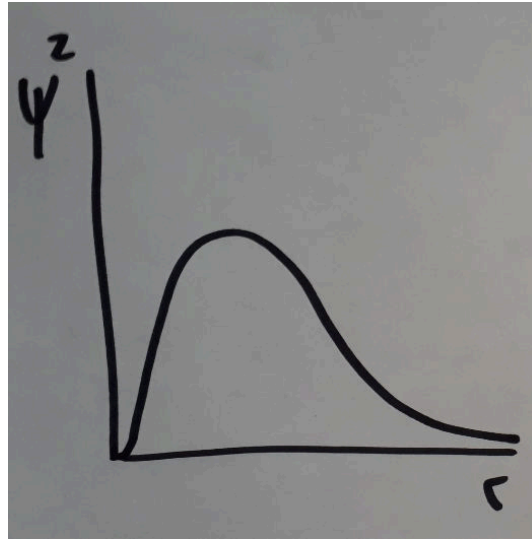
c) 4s

## Example: Sketching Radial Probability

Sketch the radial probability of a  $4d_{xy}$  orbital starting at  $r=0$

## Example: Identifying Orbitals from Probability Plots

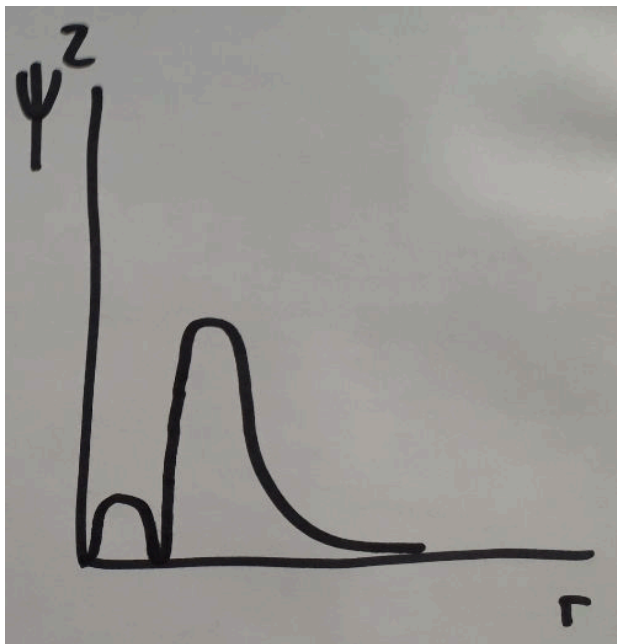
Orbital X has 2 angular nodes and its radial probability plot is shown below. Which of the following orbitals is most likely orbital X?



- a)  $3d_{z^2}$
- b)  $1s$
- c)  $4p_x$
- d)  $3s$
- e)  $2p_y$

## Practice: Identifying Orbitals from Probability Plots

An orbital has only one angular node which lies along the yz plane. The radial probability plot is shown below, please identify the orbital.



☐  $4d_{z^2}$

☐  $4d_{xy}$

☐  $3p_z$

☐  $2s$

☐  $3p_x$